Review_Insights.pdf

GenAI Case Study: Customer Sentiment & Review Response Automation
Milestone 1 — Review Understanding & Insight Generation

Business Question: "What are customers actually feeling and talking about?"

Dataset: 3,000 customer reviews (rating, review_title, review_text)
Prepared by: Dev

Executive Summary

Our negative reviews clearly show customers are most frustrated with **poor product quality**, as items frequently arrive defective, broken, or fail quickly due to cheap materials. While some delivery and service complaints exist, the overwhelming problem is the product itself. The support team should prioritize **expediting refunds/replacements** for faulty items and **rigorously escalating product quality feedback** to development to address these root causes. This will directly improve customer satisfaction and reduce churn.

(This summary was generated by Gemini based on 25 sample negative reviews and aggregated sentiment statistics — see Section 5 for the full GenAI analysis.)

Metric	Value
Total reviews analyzed	3,000
Positive sentiment (text-based)	1,769 (59.0%)
Negative sentiment (text-based)	779 (26.0%)
Mixed sentiment (text-based)	452 (15.1%)
Agreement: star rating vs. text sentiment	55.2%
3-star reviews that are actually text-Positive	781 of 1,000 (78%)

1. Rating Distribution & Review Length

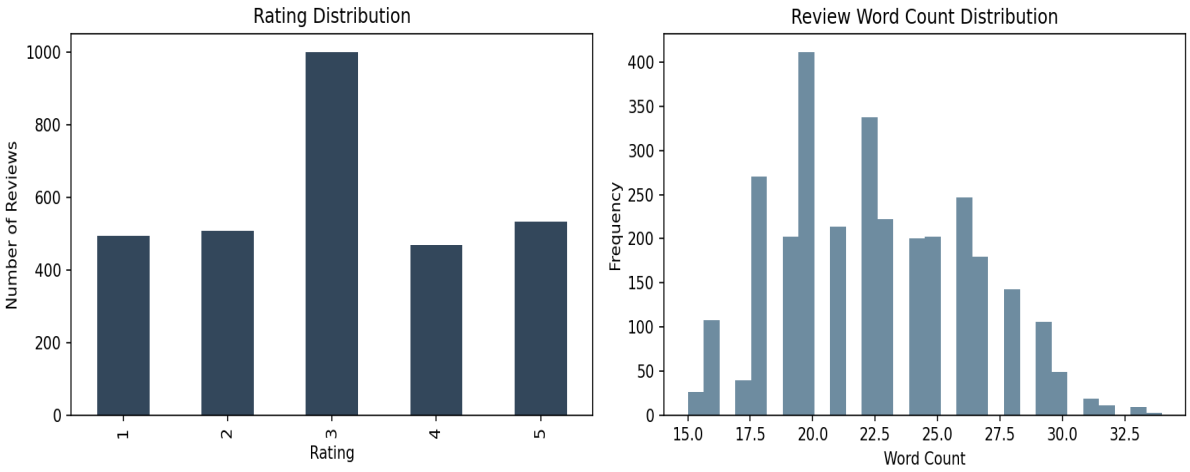


Figure 1: Rating distribution (left) and review word count distribution (right).

Key Findings

- Ratings show a strong neutral spike: exactly 1,000 of 3,000 reviews (33.33%) are rated 3 stars — more than double any other individual rating. The rest split fairly evenly: 1★ 493 (16.43%), 2★ 507 (16.90%), 4★ 468 (15.60%), 5★ 532 (17.73%).
- Review length is tightly clustered between ~15 and 34 words (mean 22.6 words / 142.1 characters), with visible peaks around 19–20 and 22–23 words rather than a smooth distribution — a signal the dataset is templated rather than fully organic, worth noting as a limitation when generalizing findings.
- Average review length by rating shows a mild but real pattern: happier customers write slightly longer reviews (5★ averages 24.5 words / 156.0 characters) while 3-star reviews are the shortest on average (21.1 words / 127.4 characters) — reinforcing that the neutral segment carries the least written detail as well as the least signal from its rating alone.

Average Review Length by Rating

Rating	Avg. Characters	Avg. Word Count
1★	143.9	22.2
2★	144.2	22.4
3★	127.4	21.1
4★	153.3	24.1
5★	156.0	24.5

Conceptual Task 1: Why Ratings Alone Are Insufficient

A 3-star rating alone tells us almost nothing — it could mean “mildly satisfied,” “average product, no strong opinion,” or mask a specific complaint buried in the text. The 3-star bucket is the biggest blind spot in this dataset and the one most worth mining with sentiment and theme analysis rather than trusting the number alone.

2. Sentiment Analysis

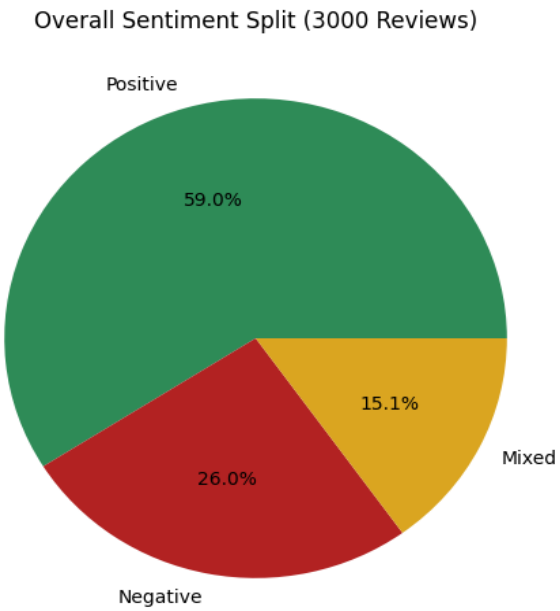


Figure 2: Overall sentiment split across all 3,000 reviews.

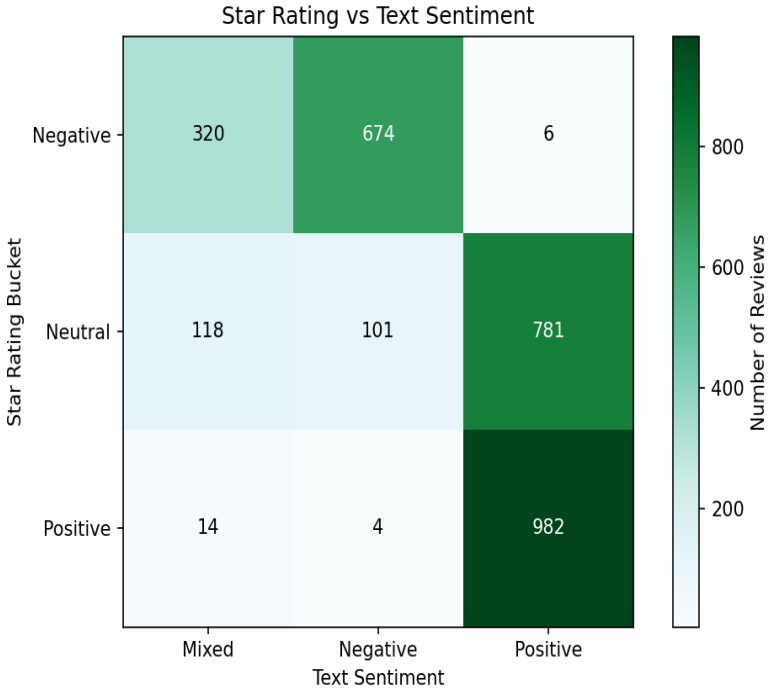


Figure 3: Star rating bucket vs. text-based sentiment.

Rating Bucket	Mixed	Negative	Positive
Negative (1-2★)	320	674	6
Neutral (3★)	118	101	781
Positive (4-5★)	14	4	982

Conceptual Task 2: Interpreting Sentiment Patterns (Business Perspective)

• Overall, text-based sentiment and star ratings agree only 55.2% of the time — nearly half of all reviews say something different than their star rating suggests.

• The biggest surprise is the Neutral (3★) segment: 781 of 1,000 three-star reviews (78%) actually read as Positive in their text. Only 101 are genuinely negative in tone. The 3-star bucket, often deprioritized by support teams as “average,” actually hides a mostly-satisfied population plus a smaller but real subset of genuinely unhappy customers who could be missed if only 1-2★ reviews are escalated.

• Within the Negative (1-2★) rating group, 674 reviews (67%) are confirmed negative in tone and 320 (32%) are Mixed — almost all low-rating customers are expressing real frustration.

• Recommendation: Support teams should combine rating + text sentiment for triage, since sentiment analysis catches unhappy customers hiding in the neutral/mid-range ratings that a rating-only filter would miss.

3. Keyword & Theme Extraction

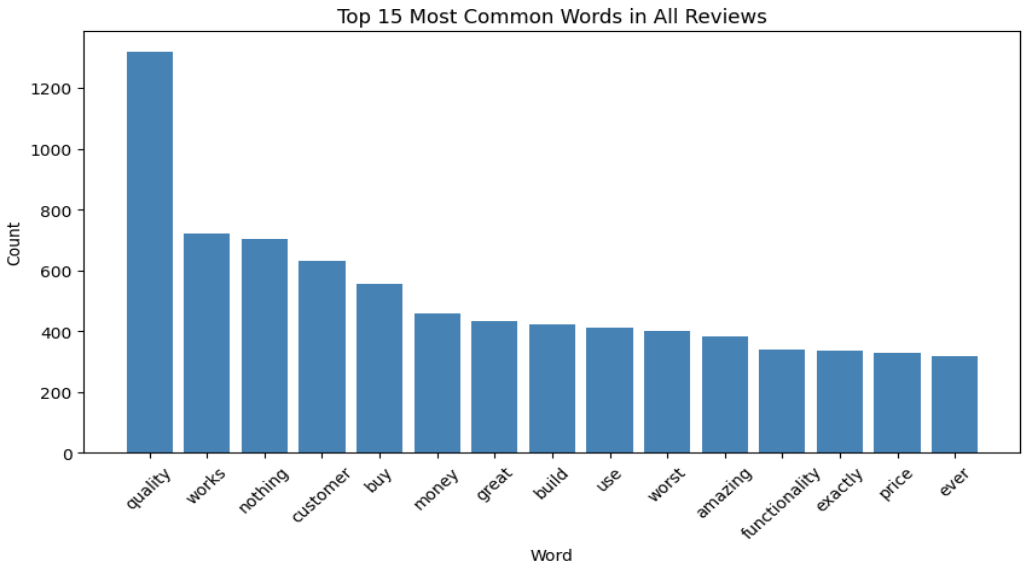


Figure 4: Top 15 most common words across all reviews.

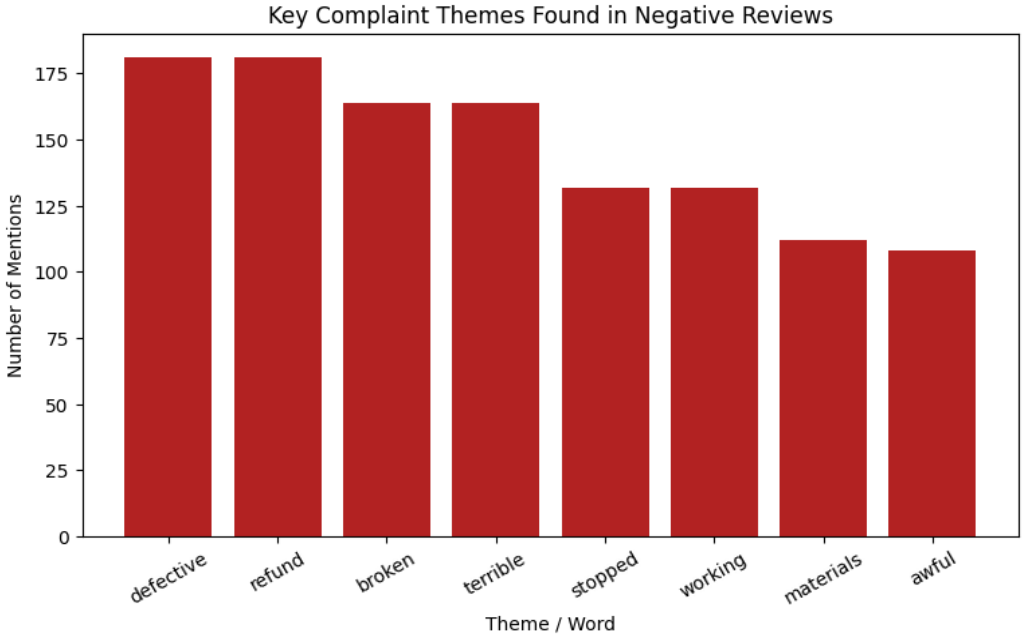


Figure 5: Key complaint themes found specifically in negative reviews.

Conceptual Task 3: Themes Indicating Serious Customer Dissatisfaction

• Product failure language: “defective” (181), “broken” (164), “stopped / working” (132) — indicates the product physically failed, not just a matter of taste.

- Resolution-seeking language: “refund” (181), “immediately” (165), “want” (149) — indicates the customer expects a specific business action, not just venting.
- Strong negative descriptors: “worst” (272), “terrible” (164), “awful” (108) — high-intensity dissatisfaction as opposed to mild disappointment.
- Generic words like “quality,” “customer,” “nothing,” and “build” appear in both positive and negative reviews and are not reliable standalone signals — a limitation of simple keyword counting that GenAI-based summarization helps address.

Conceptual Task 4: Emotional Language vs. Actual Product Issues

Not every negative-sounding review points to a real, fixable problem. **Emotional/tone language** (“terrible,” “awful,” “worst”) communicates strong dissatisfaction but no actionable detail on its own. **Actual product issue language** (“defective,” “broken,” “stopped working,” “materials”) describes a concrete, reproducible failure that should trigger real investigation (e.g. checking a batch or supplier). A third category — direct requests such as “refund” — should be treated as highest priority, since the customer has already stated the resolution they want. A response system should treat these three categories differently: empathy for tone-only language, operational follow-up for concrete issues, and immediate action for direct requests.

4. GenAI-Generated Business Insights

To go beyond keyword counts, Gemini was given a random sample of 25 negative reviews plus the aggregated sentiment/keyword statistics above, and asked four targeted business questions.

Q1: What are customers most frustrated about?

Customers are most frustrated by products that are defective or broken upon arrival, or that fail very quickly after purchase. This is consistently linked to poor build quality, cheap materials, and products not working as advertised, leading to a strong demand for refunds.

Q2: Product quality issues, delivery/service issues, or both?

Overwhelmingly product quality issues. While there are a few mentions of items arriving damaged during delivery and unhelpful customer service, the vast majority of complaints center on products being defective, broken, poorly constructed, using cheap materials, or stopping working prematurely.

Q3: What should the support team prioritize fixing first, and why?

- Efficiently addressing and resolving product defect/failure complaints with refunds or replacements — many reviews explicitly demand a refund, indicating this is the immediate relief customers seek.
- Systematically collecting and escalating detailed product quality feedback to product development and quality assurance teams, to prevent recurrence of the same defects.

Q4: 10-second manager summary

Our negative reviews clearly show customers are most frustrated with poor product quality, as items frequently arrive defective, broken, or fail quickly due to cheap materials. While some delivery and service complaints exist, the overwhelming problem is the product itself. The support team should prioritize expediting refunds/replacements for faulty items and rigorously escalating product quality feedback to development to address these root causes. This will directly improve customer satisfaction and reduce churn.

Conceptual Task 5: How GenAI Helps Summarize Large Volumes of Text

GenAI helps summarize large volumes of text by reading many individual reviews (or reasoning over pre-aggregated stats) and synthesizing them into a few coherent, human-readable sentences — something that would take a person hours to do manually across thousands of reviews. Instead of manually reading all 779 negative reviews, Gemini was given a representative sample of 25 negative reviews plus already-computed statistics, and asked targeted business questions. In seconds, it produced a structured, prioritized answer: identifying that complaints are overwhelmingly about product quality (not delivery), separating root causes from symptoms, and recommending concrete next steps. This demonstrates GenAI's core summarization strength — it doesn't just count words like keyword frequency analysis does, it understands context, groups related ideas together, and produces a summary a manager can read in 10 seconds.

5. Summary & Recommendations

• Do not rely on star rating alone for triage — 45% of reviews show a mismatch between rating and actual sentiment, and the 3-star bucket in particular hides both mostly-satisfied customers and a smaller genuinely unhappy segment.

• Prioritize reviews containing resolution-seeking and product-failure language (“refund,” “defective,” “broken,” “stopped working”) for fastest escalation — these represent concrete, actionable complaints rather than general venting.

• The root cause of dissatisfaction is overwhelmingly product quality, not delivery or service — feedback should be systematically routed to product/QA teams, not just resolved case-by-case in support.

• GenAI (used here via Gemini) is effective for turning large volumes of review text into concise, prioritized, business-readable summaries, complementing (not replacing) quantitative sentiment and keyword analysis.

Note: This dataset shows tightly clustered, templated review lengths and phrasing, suggesting synthetic generation. Findings demonstrate the analytical technique effectively but may not fully generalize to organic real-world review data.